

Assignment

- Design and implement the control unit of a washing machine that is described as follows:

The machine can be controlled by three buttons (STOP, RUN, PAUSE).

The machine has two LEDs including Red LED (RLED) and Blue one (BLED).

- RLED is on when machine is on standby (i.e., it is available to serve)
- RLED is blinking if the machine has errors
- BLED is on when the machine is ready to execute
- BLED is blinking when the machine is running (do washing)

The washing machine is only able to execute once at least 50-cents is added.

The machine only accepts 10-cent, 20-cent, 50-cent coins.

When there is enough money (BLED is on), the machine will execute if the RUN button is pressed. Then, a **30-mins timer will be activated** and the money will be clear without returning the redundancies (if any).

When the machine is running (BLED is blinking), it will be paused as the PAUSE button is pressed but **the timer is still counting down**. The machine will re-execute when the RUN button is pressed again.

The executing machine will be automatically terminated after 30 minutes (washing cycle) or when the STOP button is **pressed twice** (force to stop).